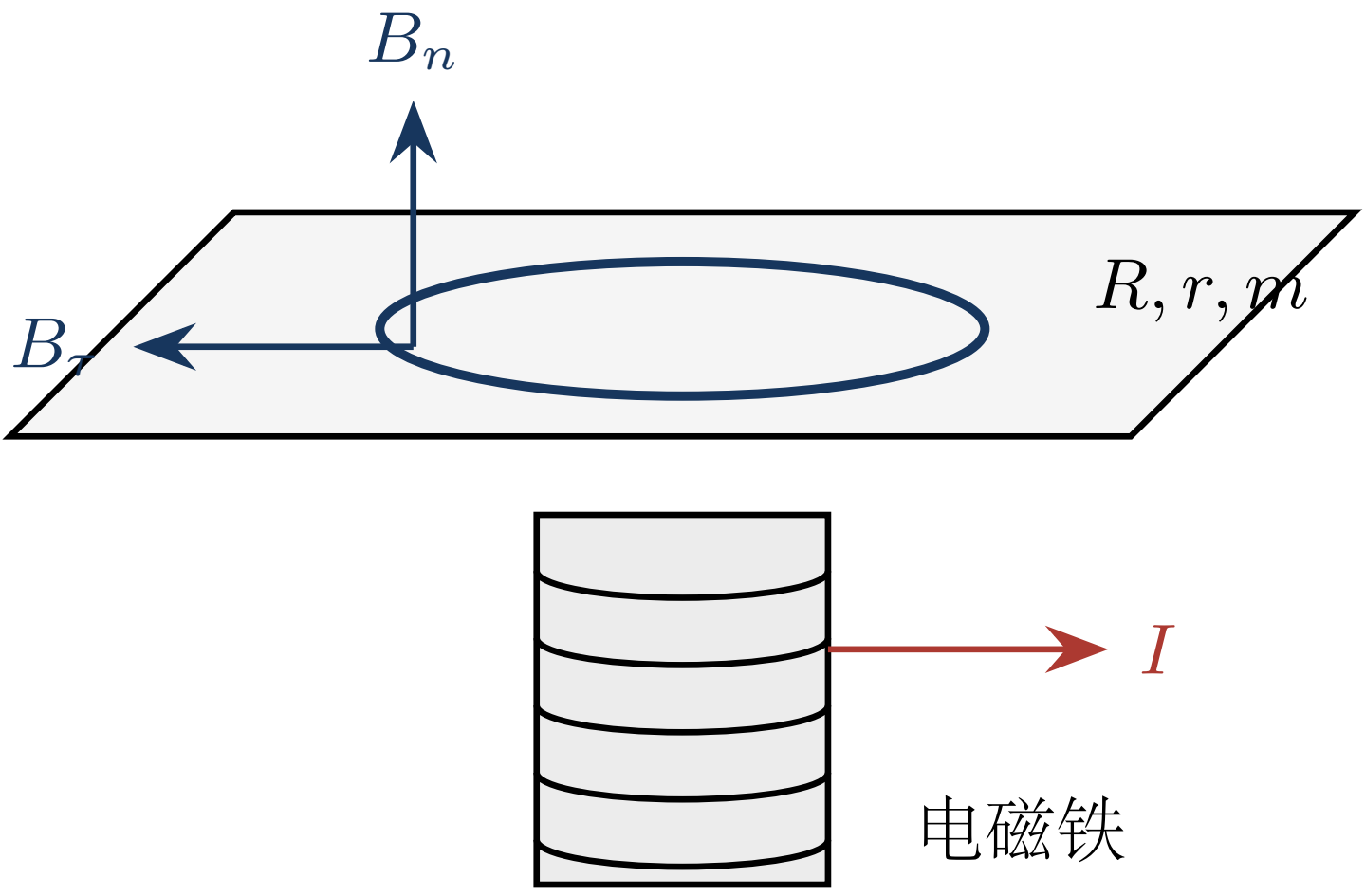
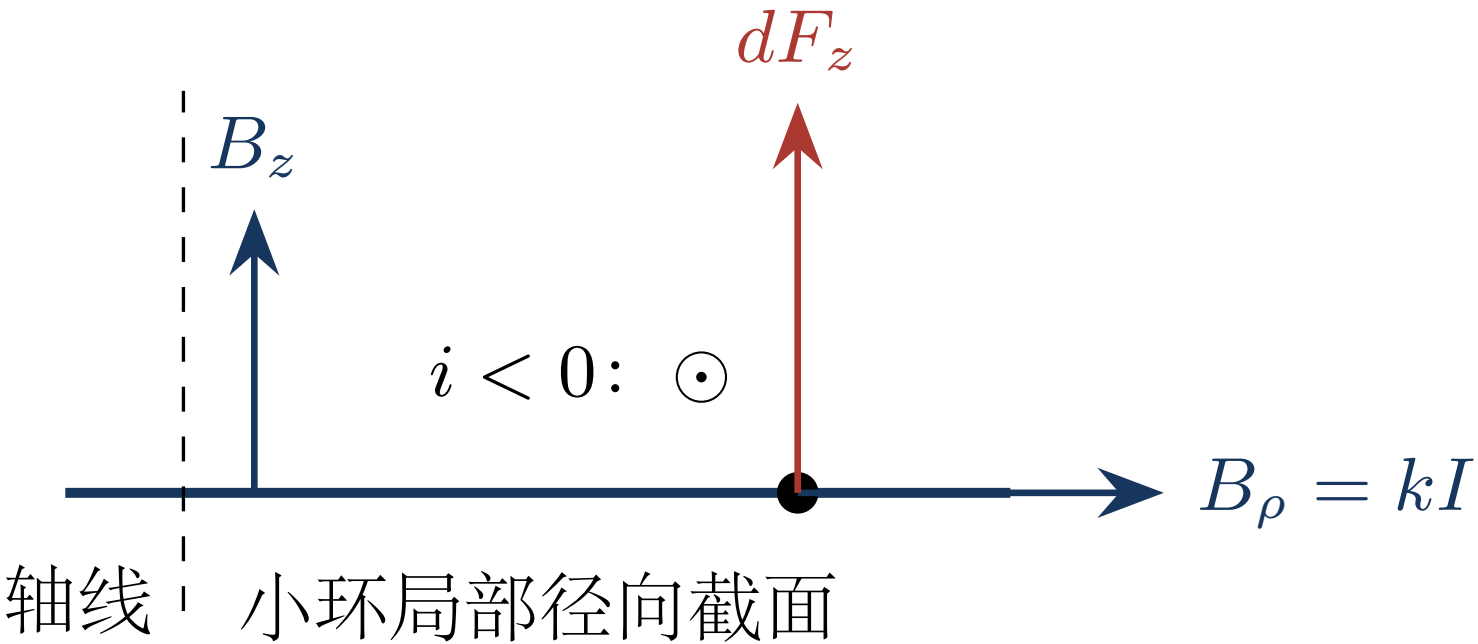
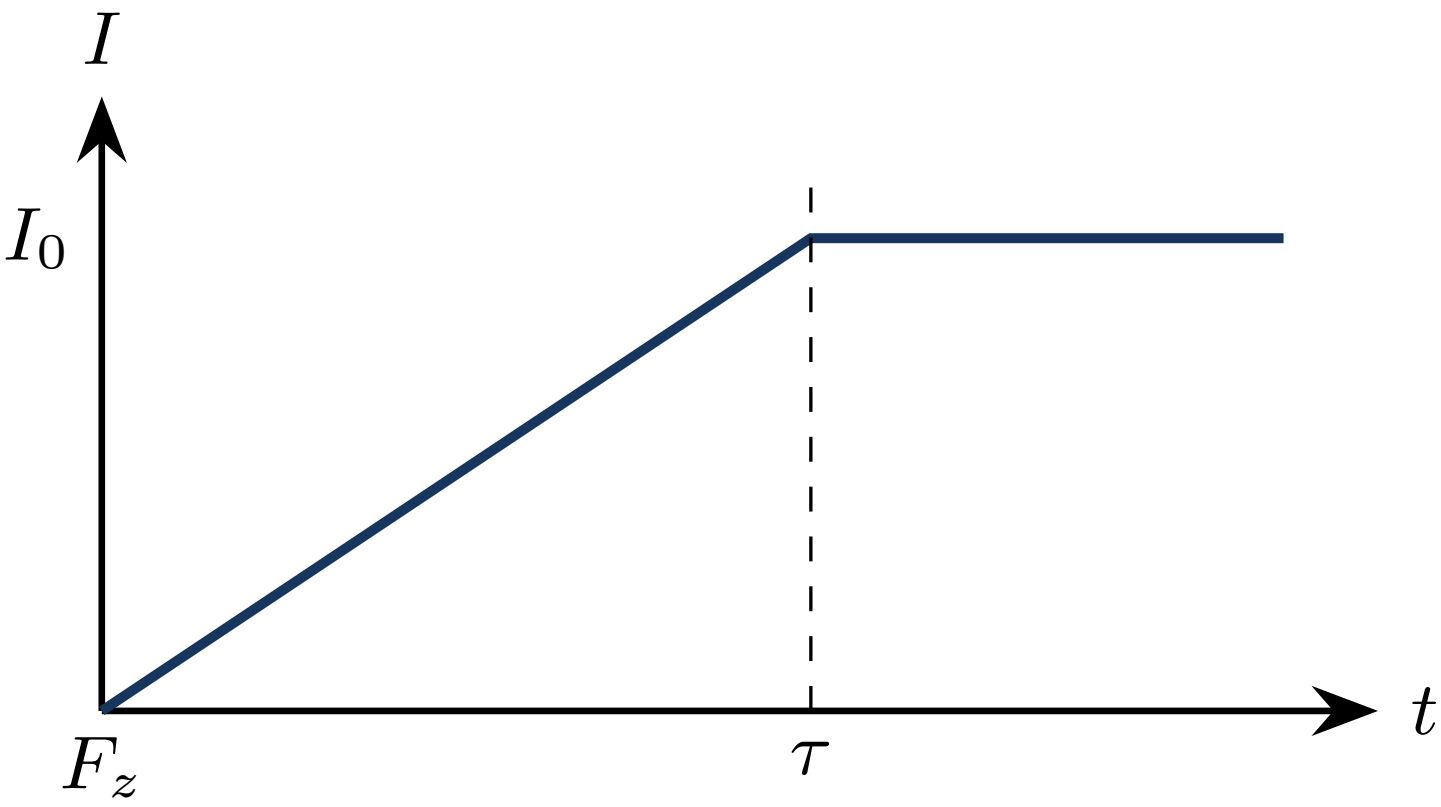
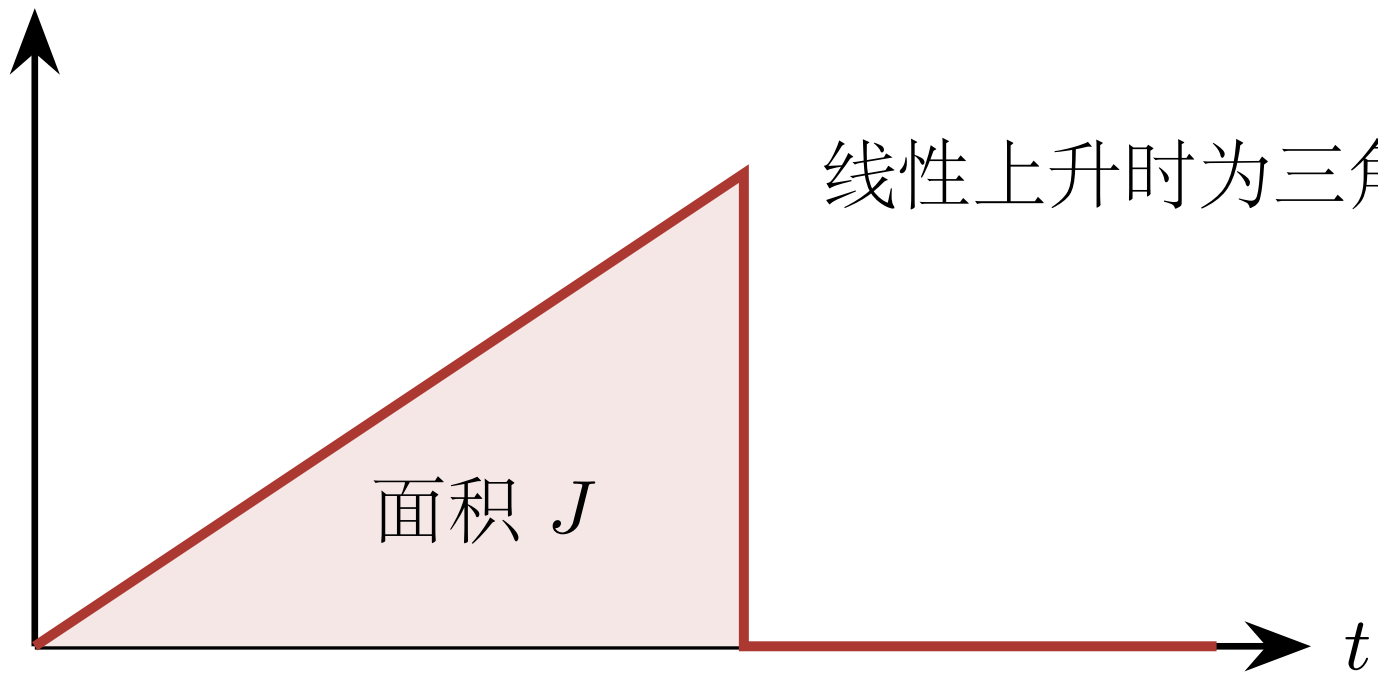

B_n
B_τ
R, r, m
I
电磁铁

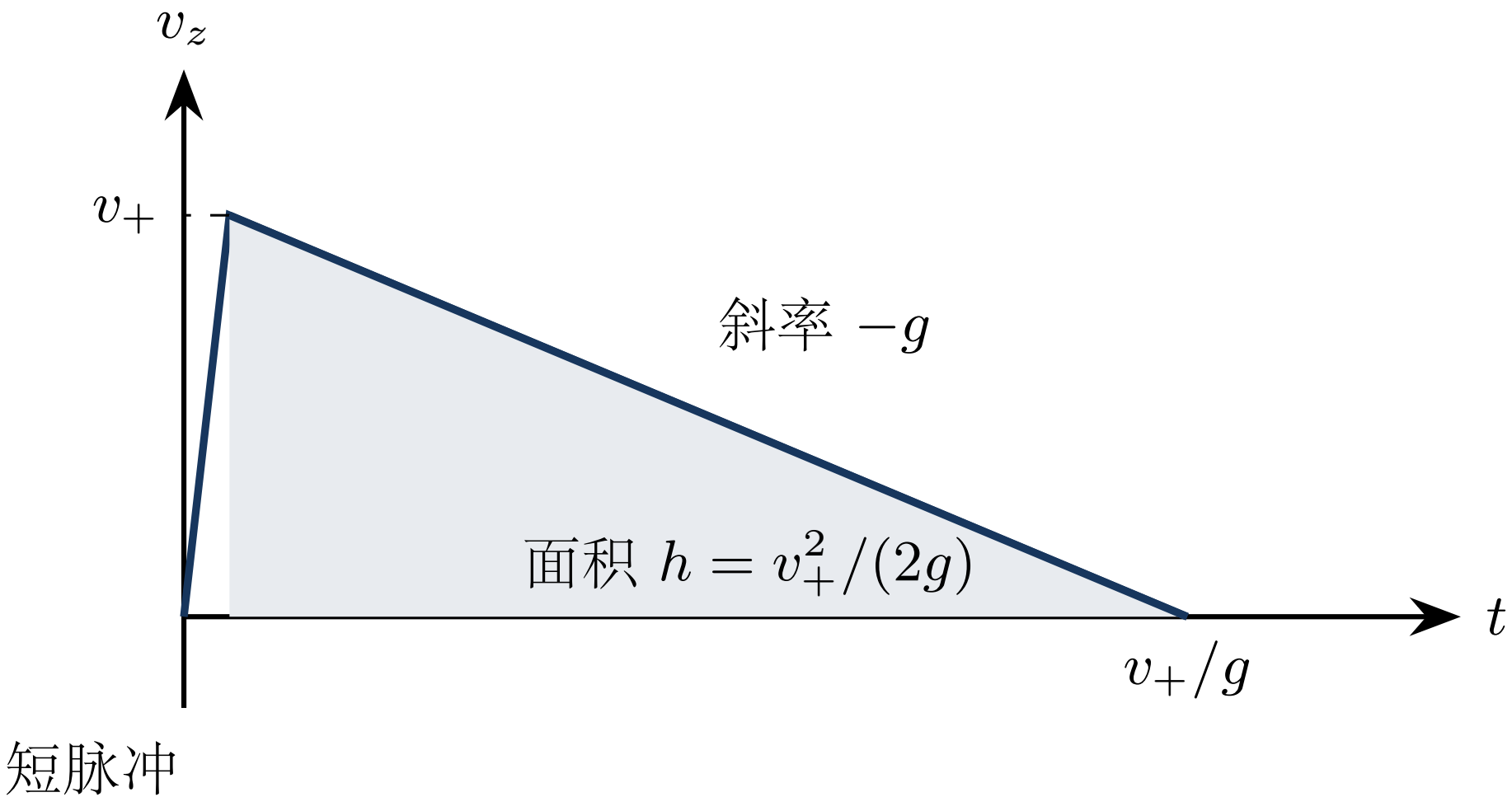


$$F_z = -2\pi RkIi$$





线性上升时为三角脉冲



短脉冲

$\mathcal{L}/r \ll \tau$

连续上升、忽略自感

↓

$J = KMI_0^2/(2r)$

$h = J^2/(2m^2g)$

$\tau \ll \mathcal{L}/r$

先取严格阶跃

↓

$J_{\text{step}} = KMI_0^2/r$

$h_{\text{step}} = 4h$

两种情况下电磁脉冲均须远短于机械时间